# COMP 3500SEF · Final Report

**Warehouse Logistics Management System**
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## 1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The application uses Next.js, TypeScript, Prisma and SQLite. GPS, scanning, telephony, payment and carrier integration are outside scope. Deployment is local-only, serving a single warehouse and the four roles above; transactional guards — stock checks and one confirmation transaction (§5) — enforce data integrity.

## 2. Development process

**Table A. Increment, trigger and outcome**

| Increment and date | Trigger | Outcome |
|---|---|---|
| Planning, 9 and 11 Oct 2026 | Project selected | Scope, responsibilities and contracts recorded (increment register) |
| 17 Oct 2026 review | Order and stock functions complete in development | Defects recorded in the increment register; led to the confirmation transaction (§5) |
| Confirmation, 26–28 Oct 2026 | Review found a failing confirmation endpoint and a missing ledger | Transaction verified (orders tests) |
| Integrated application, 1 Nov 2026 | All four role workflows runnable together | All four role workflows verified in the 11 Nov regression (browser checks) |
| Driver and privacy, 2 Nov 2026 | Driver workflow and data minimisation outstanding | Driver assignment and data-minimisation checks pass (tests) |
| Documentation hardening, 10–11 Nov 2026 | Review found diagrams and traceability diverging from the implementation | Diagrams and traceability aligned (traceability) |
| Export simplification, 11 Nov 2026 | Admin export took four interactions (issue #15) | Selected-date CSV export in three interface interactions (walkthrough record) |
| CI health tests, 11–12 Nov 2026 | The CI health record had no regression test | Two regression tests added; suite reached 38 (unit.log) |

## 3. Requirements

### Functional requirements and verification

| FR | The system shall | Verification |
|---|---|---|
| FR-01 | create orders with product lines and totals | Test — creation and stock |
| FR-02 | enforce lifecycle and stock reservation or release | Test — lifecycle and stock |
| FR-03 | number orders and shipments and show owner tracking (the staff member or driver currently holding each order) | Test — identifiers and tracking |
| FR-04 | show inventory, replenishment, ledger (inventory movement log) and low-stock alerts | Test — inventory and alerts |
| FR-05 | record event notifications and allow reading | Test — notifications and reads |
| FR-06 | provide admin aggregates and CSV export | Test — aggregates and export |
| FR-07 | authenticate and enforce role and object access | Test — authentication and boundaries |
| FR-08 | assign shipped orders and simulate delivery | Test — assignment and delivery |

NFR-01 performance (API p95 ≤2 s, first contentful paint ≤3 s); NFR-02 CI health (green build, test and browser runs recorded per increment); NFR-03 least privilege and data minimisation — role gates on every mutating endpoint; staff responses omit customer identity and delivery address (privacy tests); NFR-04 scripted operation effort — every core operation completes in at most three interface interactions (walkthrough, §8); NFR-05 process quality — CI-verified increments (workflow).

**Table B. Risk coverage** (detailed mapping)

| Risk | Addressed by | Evidence |
|---|---|---|
| RK-01 scheduling slip | Increment register | scheduling plan |
| RK-02 contract rework | Transition tests pin the FR-02 contract | transition tests |
| RK-03 test-week capacity | Traceability | scope priorities |
| RK-04 setup delay | NFR-05 | CI workflow |
| RK-05 demo data | Seed regression test covers FR-01/04/06 | seed regression test |
| RK-06 coursework assignments and test-week squeeze | Risk plan | assessment schedule |
| RK-07 stack unfamiliarity | Setup guide and CI lower ramp-up cost | setup guide |
| RK-08 report and video compete for time | Risk plan | report and video plan |

## 4. System models

The four layers — Role-specific pages, HTTP routes and authentication, Order, inventory, driver, notification and reporting business handlers, Prisma/SQLite persistence — each isolate one concern. Figures 1–5 answer, in order: role functions; transitions and stock effects; data relationships; the confirmation transaction; layer connections.

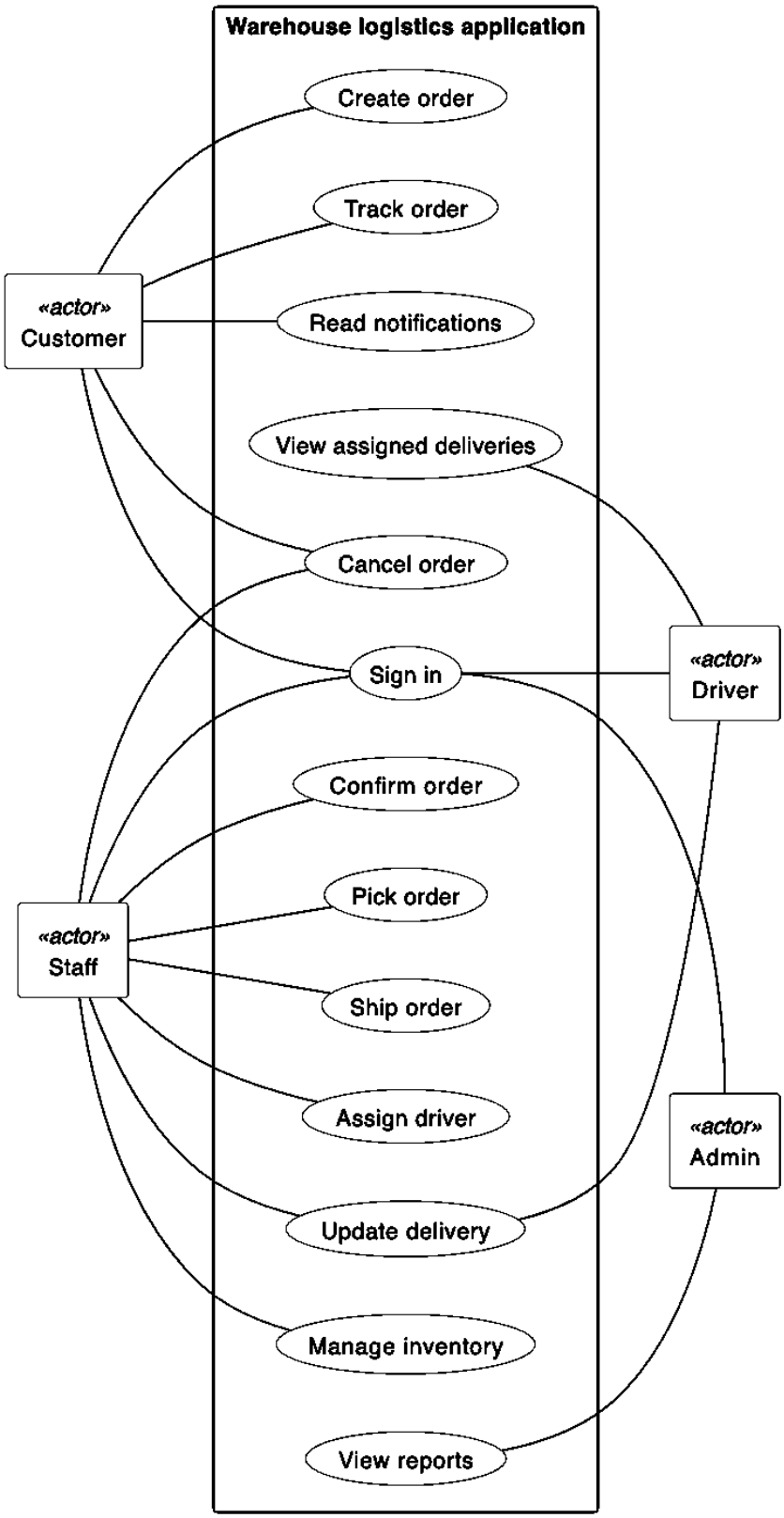



*Figure 1. Role functions.* Source.

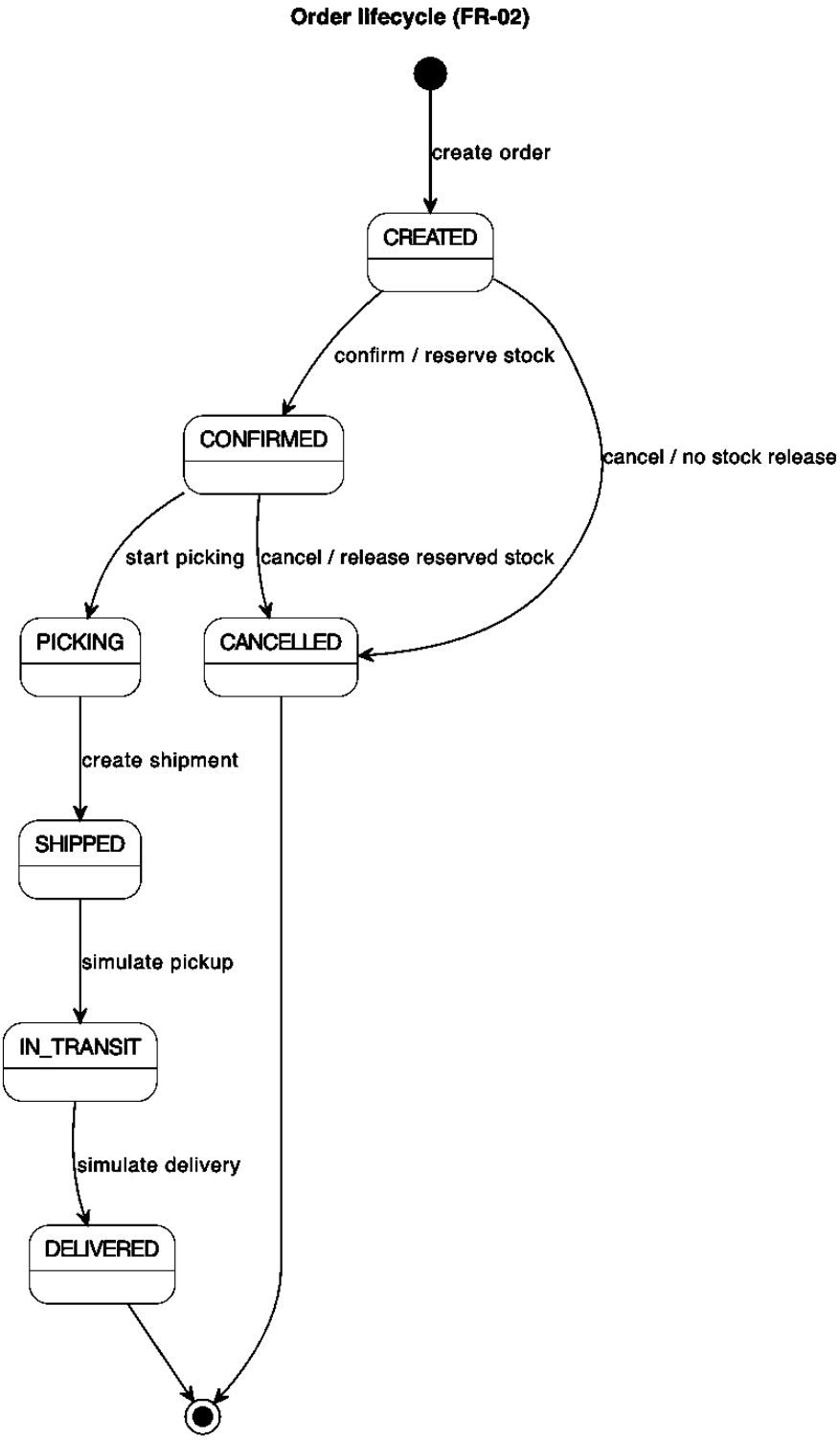



*Figure 2. Order transitions and stock effects.* Source.

Each item references one product; each order requires at least one item; orders have 0..1 shipment; products have 0..* ledger entries; users receive notifications, and each notification relates to at most one order.

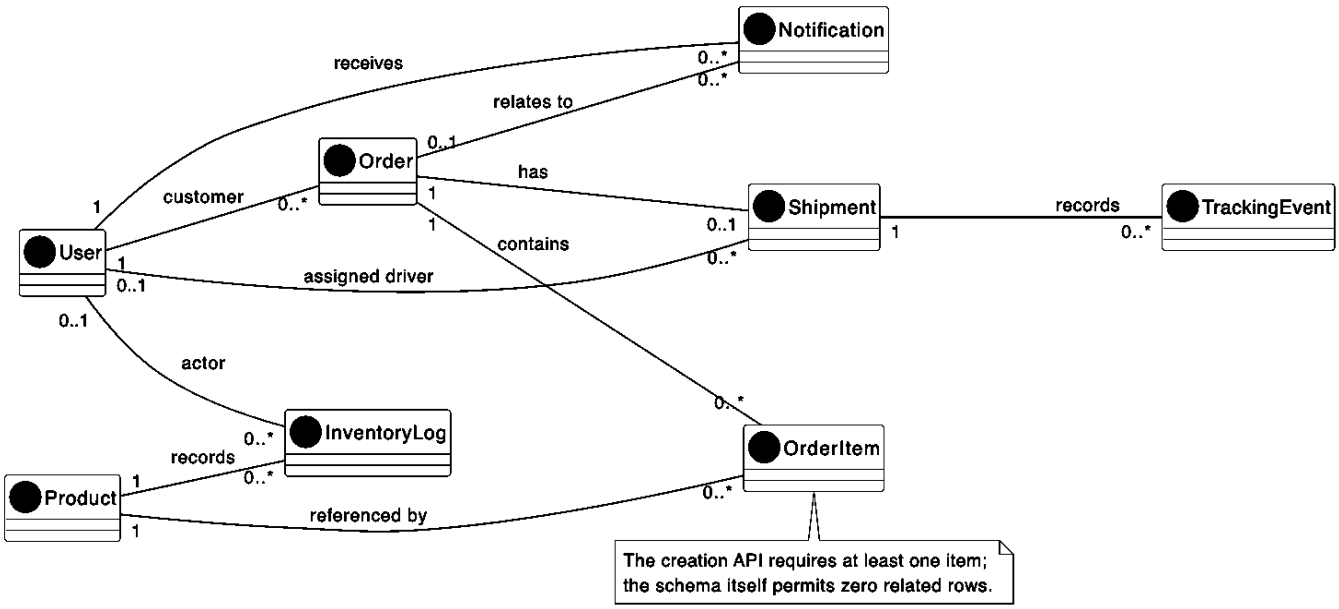



*Figure 3. Data relationships.* Source.

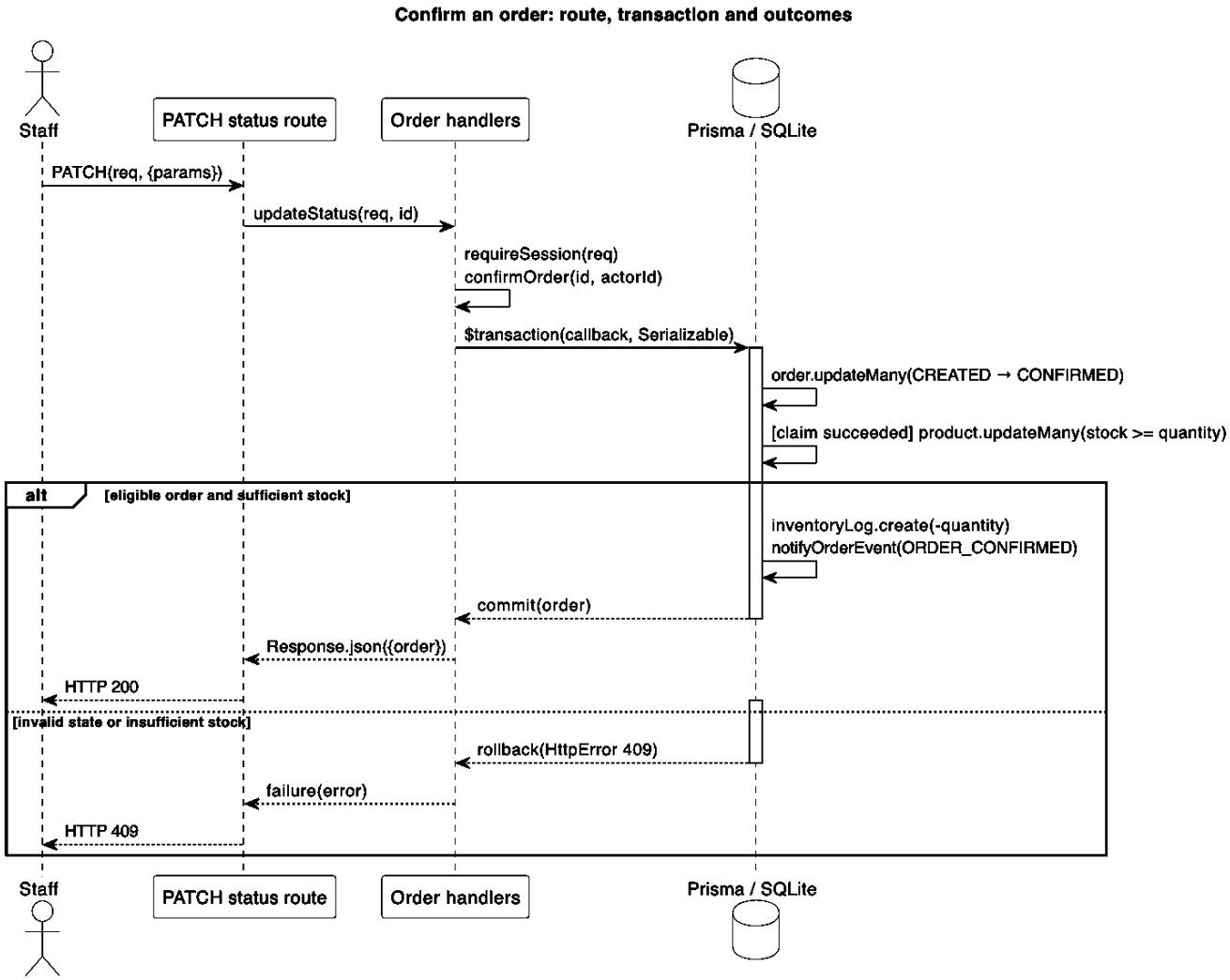



*Figure 4. Confirmation transaction.* Source.

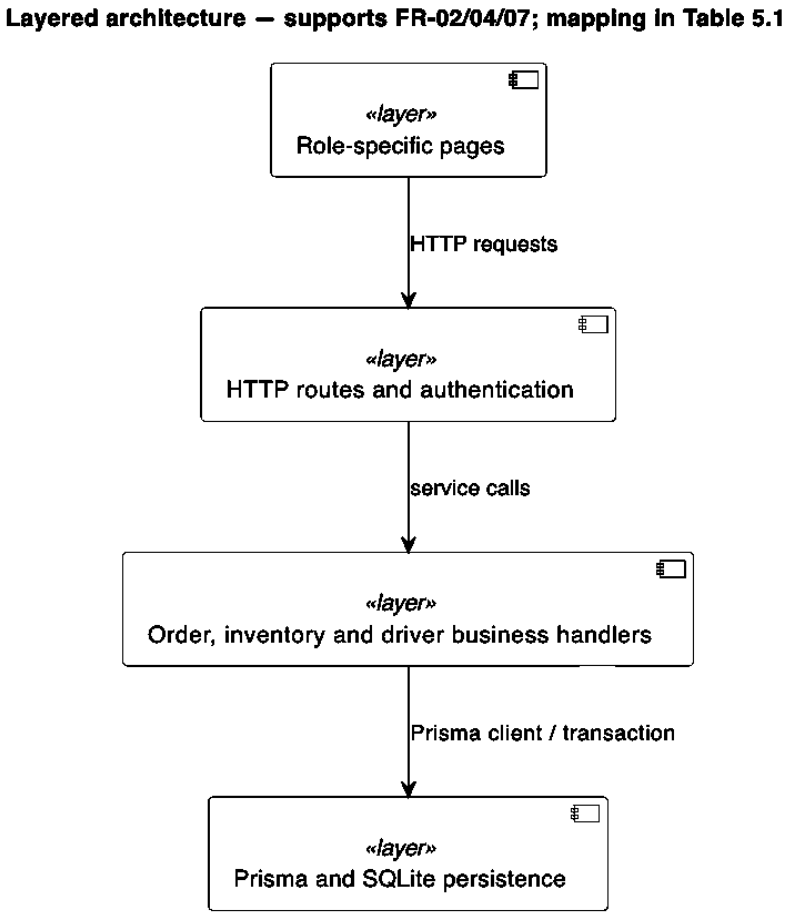



*Figure 5. Four layers.* Source.

## 5. Implementation

Stock is checked at creation and reserved at confirmation; the confirmation transaction (order-api.ts:245–293) enforces both. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED, with CANCELLED reachable from CREATED or CONFIRMED; cancellation per the matrix below. Shipments are created when staff ship.

| Role | State | Effect |
|---|---|---|
| Customer | CREATED | cancel, no restock |
| Customer | CONFIRMED | cancel, restock |
| Staff | CONFIRMED | cancel, restock |
| Customer, staff | PICKING onwards | rejected |

| Design | Implementation | Verification |
|---|---|---|
| Guards: CREATED and sufficient stock | order-api.ts:250–253, 276–280 | Orders |
| Confirmation transaction: single transaction writing state, stock, ledger and notifications; rolls back on insufficient stock or repeated confirmation | order-api.ts:245–293 | Orders |
| driverId: assign only SHIPPED; retained after delivery | driver-api.ts:44–51; transition | Driver |
| FR-03 numbering and tracking | create-order.ts:91, shipment-api.ts:62 | Logistics chain |
| FR-04 ledger | order-api.ts:281, 398 | Inventory |

## 6. Verification and validation

Verification checks requirements against the implementation; validation exercises the workflows through the interface (§8). CI records 38 unit, 140 integration and 2 delivery checks (delivery checks — packaging and evidence integrity); the regression log reads "Tests 38 passed (38)" and the walkthrough record ends "Browser acceptance passed: 26 checks".

In a local-loopback measurement (11 Nov), API p95 was 3.2–13.6 ms across endpoints and first-contentful-paint p95 was 64–96 ms, against targets of at most 2 s and 3 s. NFR-02 is evidenced by recorded green build, test and browser runs (observations).

## 7. Evolution

The 17 Oct 2026 review exercised the running app and found a confirmation endpoint that accepted repeat submits and a missing ledger; the confirmation transaction and its retest followed (26–28 Oct). The 10–11 Nov documentation review corrected the diagrams and traceability. Two CI-health regression tests added on 11 Nov brought the suite to 38 (unit.log).

## 8. Validation

The automated walkthrough exercises eleven core operations — create order; order details; confirm; pick; ship; assign; pickup; delivery; report filter; export report; selected-date export (simplified per §9) — each by its owning role, at most three interface interactions each; the walkthrough's 26 Playwright browser checks cover them at one to three checks each, with expected outcomes asserted in the script. Results are in the walkthrough record; the team reviewed them against expected outcomes.

## 9. Feedback and reconciliation

**Table C. Feedback to evidence**

| Feedback | Disposition | Source | Eviden |
|---|---|---|---|
| Diagram bases diverged from the implemented schema | Models corrected | 10–11 Nov documentation review | Diagran |
| Stock effects were recorded under FR-04 | Moved to FR-02 | 17 Oct review | Traceal |
| Trigger and outcome detail was thin | Register rows rewritten with concrete triggers and outcomes | 10–11 Nov documentation review | Registe |
| The admin export took four scripted interactions | Simplified to three interactions; verified in the record | 11 Nov automated walkthrough | Issue # |
| The CI health record had no regression test | Suite reached 38 | 11 Nov regression work | Test his |
| NFR criteria assumed a hosted deployment model | Set for local deployment | 10–11 Nov documentation review | Require |

## 10. Evidence

Traceability · Baseline record — the recorded 12 Nov suite: 38 unit, 140 integration, 2 delivery and 26 browser checks.